INCH-POUND
A-A-59198A
11 October 2002
SUPERSEDING
A-A-59198
31 March 1998

COMMERCIAL ITEM DESCRIPTION

PSYCHROMETERS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for psychrometers used for determining atmospheric relative humidity.

2. CLASSIFICATION. The psychrometers shall be classified by the following types as specified (see 7.4(b)).

Type II - Sling
Type III - Pocket sling
Type IV - Fan ventilated, battery-operated

3. SALIENT CHARACTERISTICS

3.1 Components. The psychrometer shall consist of two sturdy identical liquid-in-glass thermometers, one of which is fitted with a wick for wetting the bulb, a corrosion resistant frame, mounting clips, and a psychometric chart. Unless otherwise specified (see 7.4(c)), the thermometers shall indicate degrees Fahrenheit (ºF) with a range of 20 ºF to 120 ºF (or degrees Celsius (ºC) with a range of -6 ºC to 49 ºC) and an accuracy of ± 1 degree. When specified, spare wicks (see 7.4(d)), spare cotton thread for tying the wick (see 7.4(e)), a plastic relative humidity calculator (slide rule) (see 7.4(f)), and a National Institute of Standards and Technology Certification (see 7.4(g)) shall be provided.

3.1.1 Thermometers. The thermometers shall be annealed and have an expansion chamber for temperatures of 220 ºF to 250 ºF (104 ºC to 121 ºC) (see figure 1). Unless otherwise specified (see 7.4(h)), the liquid of the thermometers shall be colored, resistant to fading, and organic based. The liquid shall be clean, free from sediment and without separation. When specified, an additional pair of thermometers shall be furnished (see 7.4(i)).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 6685

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

3.1.2 Capillary glass. The capillary glass shall have clear division lines and numbers, as well as a white or yellow back-stripe to facilitate scale readings of the colored liquid in a clear cylindrical capillary bore. The lines shall extend fully across the capillary bore and have a thickness of no more than 33 percent of the graduation space. The stripe shall be centered under the capillary and lines; every fifth line shall be longer than the others and every tenth line shall be numbered (see figure 1). There shall be an "F" at the higher end of the scale (or "C", if Celsius). Screw-fastened clips shall securely attach the thermometers so that their entire scale (lines and numbers) is readily visible.

3.1.3 Wick. The wick shall fit the bulb snugly and extend approximately 0.5 inch beyond the end of the thermometer. It shall be clean, bleached cotton and shall be tied tightly by thin cotton thread at the top of the bulb but loosely at the bottom.

3.1.4 Metal parts. All external metal parts shall be free from burrs and sharp edges.

3.2 Type II. The indicating scale of the thermometers shall be divided into increments of 1 degree Fahrenheit (or 1 degree Celsius). The scale shall have a minimum length of 7 inches, and its maximum length shall not exceed 10 inches. The rotatable sling handle shall have a contour or finish that makes it slip-resistant. The wet bulb shall be positioned so that its bulb and wick extend beyond the dry bulb (see figure 1).

3.3 Type III. The indicating scale of the thermometers shall be divided into increments of 1 degree Fahrenheit (or 1 degree Celsius) with a range of 40 ºF to 120 ºF (4 ºC to 49 ºC). The scale shall have a minimum length of 2.7 inches, and its maximum length shall not exceed 6 inches. The rotatable sling handle shall have a contour or finish that makes it slip-resistant. The wet bulb shall be positioned so that its bulb and wick extend beyond the dry bulb (see figure 2).

3.3.1 Carrying case. A pouch carrying case with rigid protection insert for the psychrometer shall be furnished.

3.4 Type IV. The indicating scale of the thermometers shall be divided into increments of 1 degree Fahrenheit (or 1 degree Celsius). The scale shall have a minimum length of 4.25 inches, and its maximum length shall not exceed 9.5 inches (see figure 4).

3.4.1 Thermometer mounting. Each thermometer shall be mounted in a recess in a heat-reflective holder module and have a resilient sleeve at each end.

3.4.2 Thermometer bulb. Each thermometer bulb shall be surrounded by an easily detachable, seamless, heat reflective, thermally non-conductive intake shield, with air to be drawn through each duct only from the fan at a speed between 15 and 32 feet per second from the end of the psychrometer furthest from the operator. The duct mouths of the shield assembly shall be dimensioned for minimal mass, consistent with sturdiness and air flow, and surround the bulbs and the extended wick.

3.4.3 Wick access. There must be free access to the wet-bulb's wick by the operator after removing the shield assembly for quickly and easily wetting the wick or periodically replacing the wick, without the need of tools or removing the thermometer from its mount.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

3.4.4 Thermometer replacement. The thermometers should be able to be removed quickly and easily for replacement, without the need for special tools.

3.4.5 Fan motor/illuminating lamp switch. The switch that turns the fan motor on and off shall be easily manipulated and shall allow the brightness of the red scale-illuminating lamp to be varied from full intensity to low intensity or the lamp turned off. Stray light shall not be emitted from the psychrometer.

3.4.6 Psychrometer housing. The psychrometer housing, with its thermometer holder-module mounted at its top, shall not exceed 10 inches in length by 4.5 inches in height by 2 inches in width, should be ruggedly constructed and have easily accessible compartments for three D-size batteries and a plastic water container (retained by a sliding plate), and should be assembled with fasteners that can be removed with standard tools for servicing the thermometers, lamp and fan/motor assembly.

3.4.7 Batteries. The three D-size batteries (to be supplied and installed by the operator), as well as the lamp and fan/motor assembly, shall be easily replaceable without the need of any special tools or soldering equipment. All electrical components shall be securely mounted inside the closed housing. When specified, an additional red lamp shall be included (see 7.4(j)) or an additional fan/motor assembly shall be included (see 7.4(k)) or both.

3.4.8 Accessories. A plastic water container, a sturdy neck/shoulder strap (with a snap clip to temporarily affix it to a fitting on the psychrometer), and an instruction manual (with numbered parts list, numbered product drawings, trouble-shooting section, circuit drawing, and percent relative humidity tables to cover the thermometer range (see 3.1) for Fahrenheit and Celsius temperatures, with depression temperatures of no more than 1 ºF (0.56 ºC)) shall be furnished. The strap provided shall also serve to manually sling the thermometers when their holder-module has been removed from the housing using standard tools.

3.4.9 Carrying case. A hinged carrying case with folding handle and compartmented resilient cushioning shall be provided to protect the psychrometer and its accessories. The case shall be able to contain the psychrometer and all the parts and instruction manual, as well as any ordered accessories.

3.4.10 Mounting bracket. When specified, a mounting bracket to easily and temporarily affix the type IV psychrometer to a wall, with its duct openings furthermost from the wall, shall be provided (see 7.4(l)).

3.4.11 Fasteners. When specified, a spare set of fasteners (screws and nuts) for the housing and thermometer mount shall be provided (see 7.4(m)).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

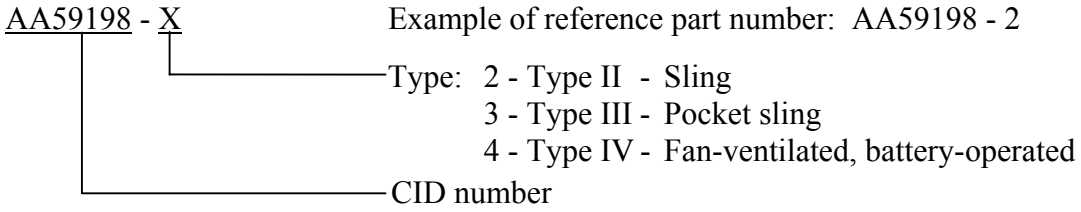
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.4(n)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 National stock number (NSN). The NSNs listed below are assigned to the listed product types. Other NSNs may also correspond with this document.

NSN	Type
6660-00-223-5083	II
6660-00-223-5084	II
6660-00-223-5089	II
6685-00-826-1662	III
6685-00-590-8759	IV
6660-00-935-1389	IV
6685-00-892-1643	IV

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products that meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

Kahl Scientific Instrument Corporation
El Cajon, CA 92022

Taylor Precision Products LP
Oak Brook, IL 60523

Weksler Instruments
Stratford, CT 06614

Bacharach, Inc.
New Kensington, PA 15068

7.4 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN.
b. Type of psychrometer required (see 2).
c. Temperature range, indicating scale-increments, and accuracy of psychrometer, if different (see 3.1, 3.2, 3.3, and 3.4).
d. A package of five wicks, if required (see 3.1).
e. A spool of cotton thread for tying the wick to the thermometer bulb, if required (see 3.1).
f. A pocket size, plastic relative humidity calculator (slide rule), if required (see 3.1).
g. National Institute of Standards and Technology Certification, if required (see 3.1).
h. Non-organic based (mercury) indicating liquid, if required (see 3.1.1).
i. A pair of additional thermometers, if required (see 3.1.1).
j. An additional red lamp, if required (see 3.4.7).
k. An additional fan/motor assembly, if required (see 3.4.7).
l. A type IV mounting bracket, if required (see 3.4.10).
m. A spare set of fasteners, if required (see 3.4.11).
n. Preservation, packing, and marking (see 6.1).

7.5 Cross reference. Table I shows the classification cross reference data.

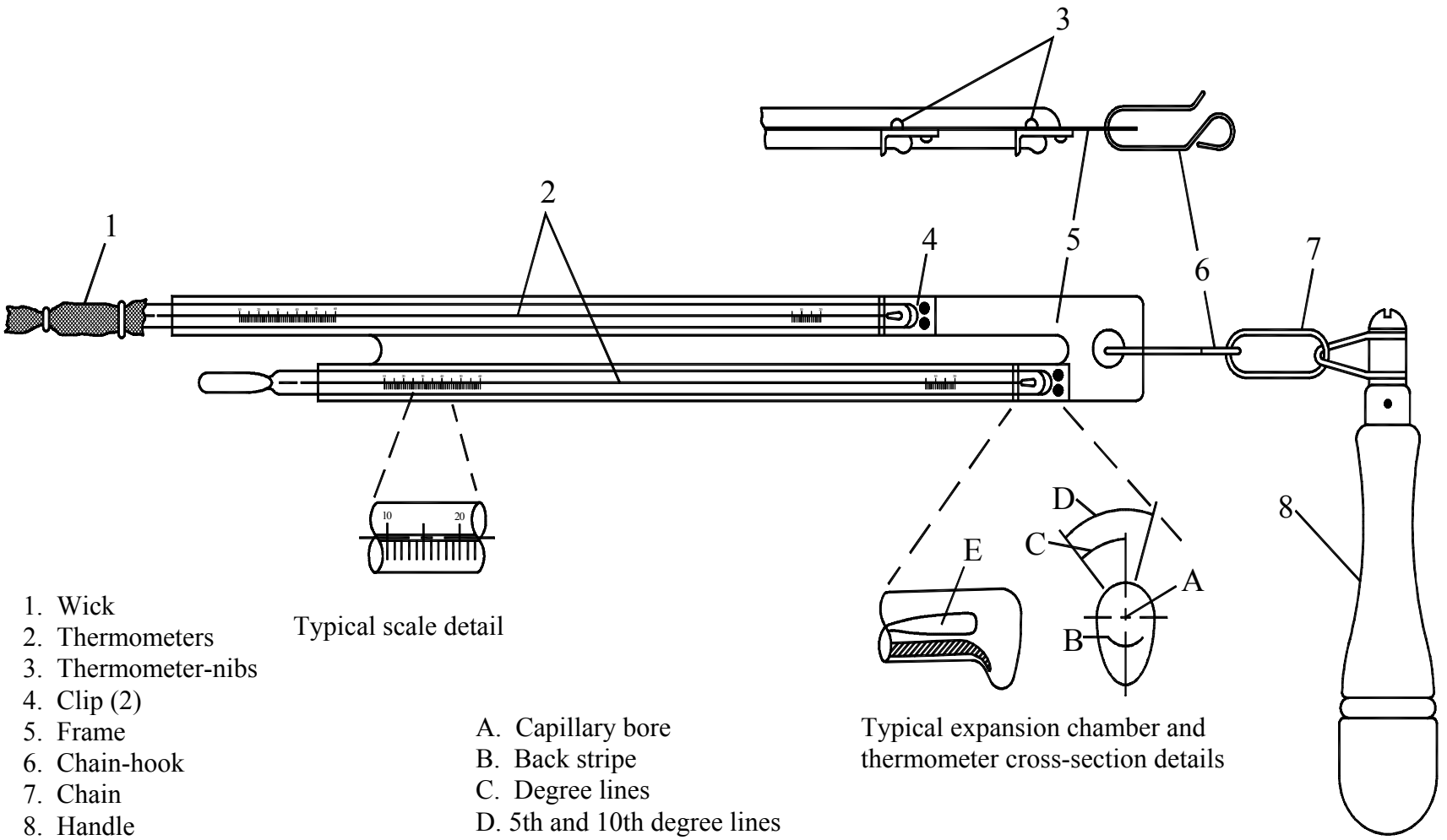
Table I. Cross reference data.

Type	A-A-59198	A-A-59198A
I	Wall mounted	Deleted
II	Sling	Sling
III	Pocket sling	Pocket sling
IV	Fan-ventilated, battery operated	Fan-ventilated, battery operated
V	Hand-held, battery operated, digital	Deleted

7.6 Subject term (key word) listing.

dry bulb
fan
pocket sling
relative humidity
thermometer
ventilated
wet bulb
wick

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.



1. Wick
2. Thermometers
3. Thermometer-nibs
4. Clip (2)
5. Frame
6. Chain-hook
7. Chain
8. Handle

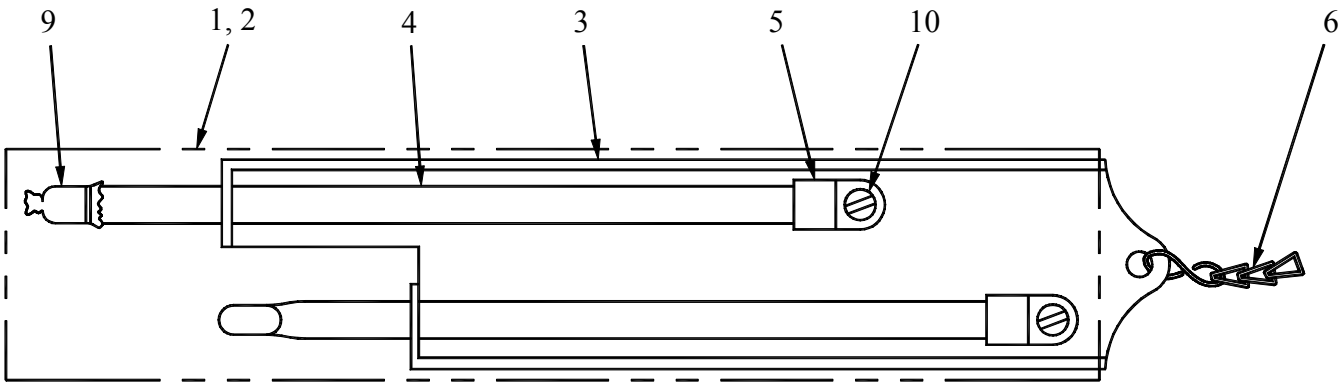
Typical scale detail

A. Capillary bore
B. Back stripe
C. Degree lines
D. 5th and 10th degree lines
E. Expansion chamber

Typical expansion chamber and thermometer cross-section details

Figure 1. Type II sling psychrometer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.



1. Case
2. Insert
3. Frame
4. Thermometer (2)
5. Clip (2)
6. Handle chain
7. Ring
8. Handle
9. Wick
10. Screw (2)

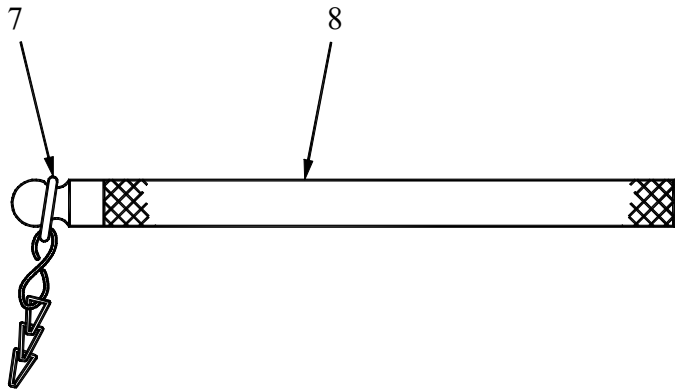


Figure 2. Type III pocket sling psychrometer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

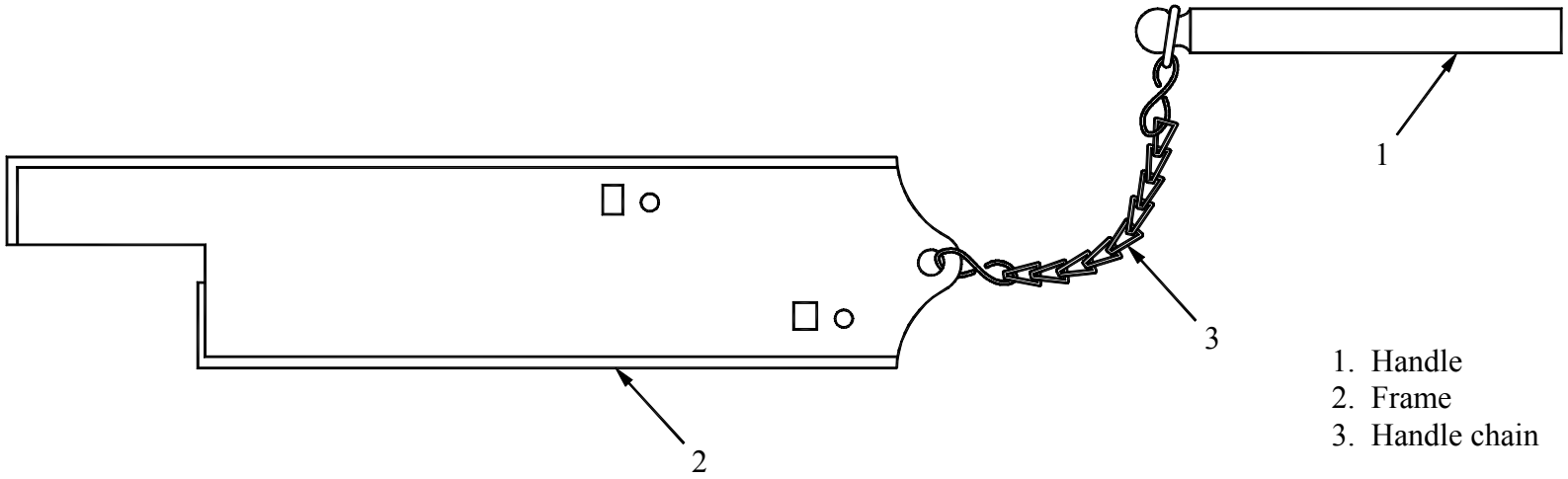
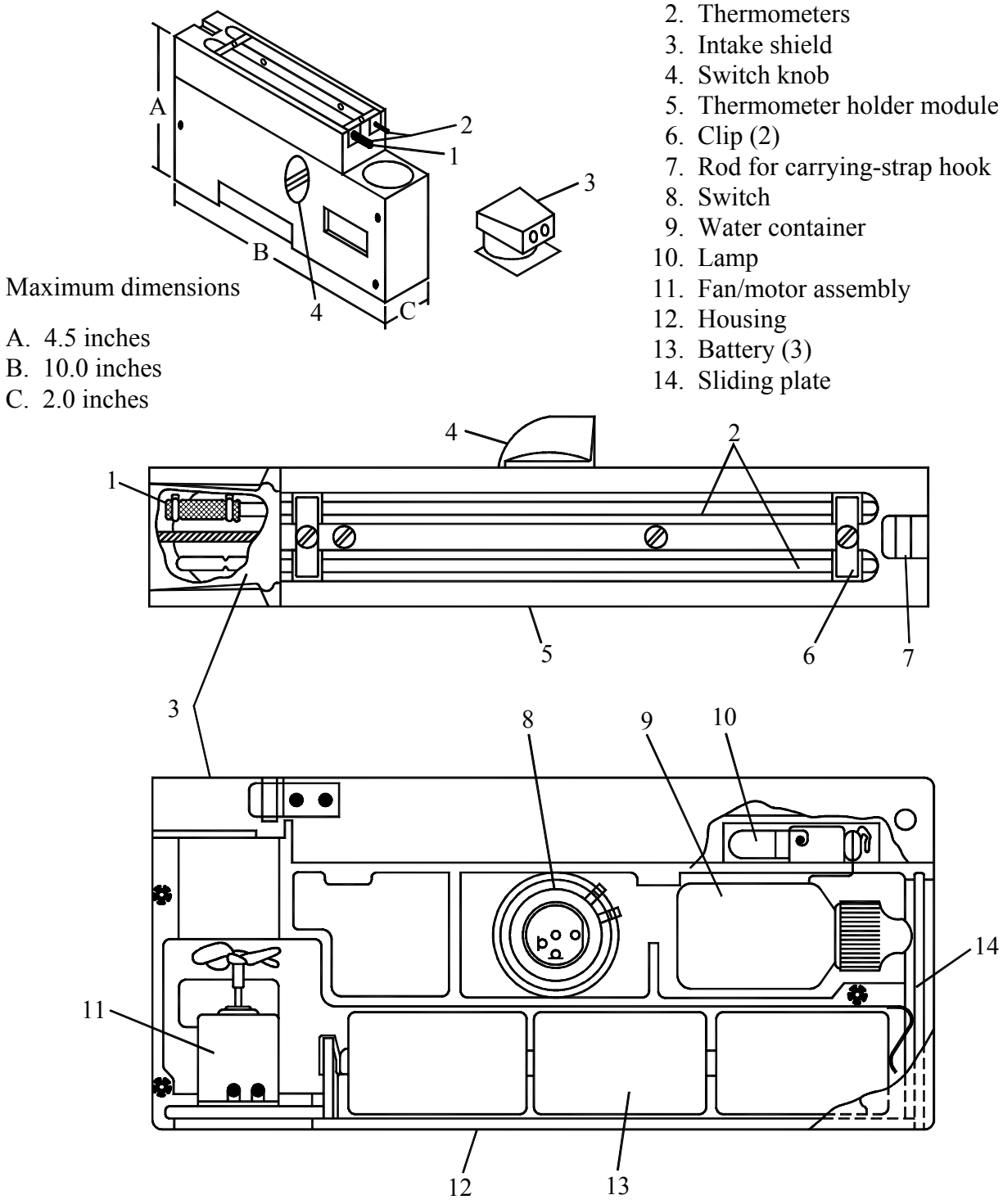


Figure 3. Type III pocket sling psychrometer frame.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.



1. Wick
2. Thermometers
3. Intake shield
4. Switch knob
5. Thermometer holder module
6. Clip (2)
7. Rod for carrying-strap hook
8. Switch
9. Water container
10. Lamp
11. Fan/motor assembly
12. Housing
13. Battery (3)
14. Sliding plate

Maximum dimensions

A. 4.5 inches
B. 10.0 inches
C. 2.0 inches

Figure 4. Type IV fan ventilated, battery-operated psychrometer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity:
DLA - GS1

(Project 6685-0064)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:05Z
Check the source to verify that this is the current version before use.